Fall 2026 DSC 242: Homework 4

Due 11:59PM Friday, October 30

Submission. Problems 1–3 are written problems. Problem 4 is an experimental problem: submit a single PDF report (for example, exported from a Jupyter notebook) that contains your figures, a short discussion for each part, and the code that produced the figures (as notebook cells or as an appendix). Every figure should have labeled axes, a legend, and a caption stating the parameters and the number of trials. You are encouraged to reuse your code from Homework 3.

Notation. Throughout, $y = X\beta^* + \epsilon$ with design $X \in \mathbb{R}^{n\times p}$, $\|\beta^*\|_0 = k$ and $S = \mathrm{supp}(\beta^*)$; the LASSO estimator is

$$\hat{\beta} = \operatorname*{argmin}_{\beta\in\mathbb{R}^p} \frac{1}{2n}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1,$$

and $\Delta = \hat{\beta} - \beta^*$. For $A \in \mathbb{R}^{n\times p}$ and $T \subseteq [p]$, $A_T \in \mathbb{R}^{n\times|T|}$ is the submatrix of the columns indexed by T, and for $u \in \mathbb{R}^p$, $u_T \in \mathbb{R}^p$ is u with the coordinates outside T set to zero.

1. (The LASSO beyond the ℓ_2 norm, 20 points) Recall the two conditions on the data (X, ϵ) from the LASSO analysis:

   (N) the noise condition $\|\frac{1}{n}X^\top\epsilon\|_\infty \le \frac{\lambda}{2}$;

   (RE) the restricted eigenvalue condition: $\frac{1}{n}\|Xr\|_2^2 \ge \kappa\|r\|_2^2$ for all $r \in \mathbb{R}^p$ with $\|r_{S^c}\|_1 \le 3\|r_S\|_1$.

   Under (N), the LASSO estimation error satisfies

   $$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{3\lambda}{2}\|\Delta_S\|_1 - \frac{\lambda}{2}\|\Delta_{S^c}\|_1, \qquad \text{in particular} \qquad \|\Delta_{S^c}\|_1 \le 3\|\Delta_S\|_1, \tag{0.1}$$

   and under (N) and (RE),

   $$\|\Delta\|_2 \le \frac{3\lambda\sqrt{k}}{\kappa}. \tag{0.2}$$

   With $\lambda \asymp \sigma\sqrt{\log p/n}$, (0.2) is the rate $\sigma\sqrt{k\log p/n}$ from the lecture.

   (a) (ℓ_1 error, 4 points) Assume (N) and (RE). Show that $\|\Delta\|_1 \le \frac{12k\lambda}{\kappa}$.

   (b) (ℓ_q error, 6 points) Show that for every $v \in \mathbb{R}^p$ and every $q \in [1, 2]$,

   $$\|v\|_q \le \|v\|_1^{2/q-1}\,\|v\|_2^{2-2/q}.$$

   Conclude that under (N) and (RE), $\|\Delta\|_q \le \frac{12k^{1/q}\lambda}{\kappa}$ for all $q \in [1, 2]$. Which values of q recover part (a) and (0.2)?

   (c) (Prediction error, fast rate, 4 points) Assume (N) and (RE). Show that

   $$\frac{1}{n}\|X\Delta\|_2^2 \le \frac{9k\lambda^2}{\kappa}.$$

(d) (Prediction error, slow rate, 6 points) Now assume *only* (N): no condition on the design, and no sparsity of β^* (any $\beta^* \in \mathbb{R}^p$, $k = p$ allowed). Show that

$$\frac{1}{n}\|X\Delta\|_2^2 \le 3\lambda\|\beta^*\|_1 \qquad \text{and} \qquad \|\hat\beta\|_1 \le 3\|\beta^*\|_1.$$

Then take $\lambda \asymp \sigma\sqrt{\log p/n}$ and a β^* with k nonzero entries equal to ± 1, and compare the prediction bounds of parts (c) and (d) as functions of n. For which n is the bound of part (c) the smaller one?

2. (Coherence and the square-root bottleneck, 20 points) Let $A \in \mathbb{R}^{n\times p}$ have columns $A_1, \dots, A_p$, and define

$$\delta_1(A) = \max_{j\in[p]} \big|\|A_j\|_2^2 - 1\big|, \qquad \mu(A) = \max_{i\neq j} |\langle A_i, A_j\rangle| \quad \text{(the } \textit{coherence}\text{)},$$

and, for $k \ge 1$, the restricted isometry constant

$$\delta_k(A) = \sup\Big\{\big|\|Au\|_2^2 - 1\big| : \ \|u\|_2 = 1,\ \|u\|_0 \le k\Big\} = \max_{|T|=k} \big\|A_T^\top A_T - I_k\big\|_{\mathrm{op}},$$

the smallest δ for which A is k-RIP with constant δ. Computing $\delta_k(A)$ requires all $\binom{p}{k}$ supports; $\delta_1(A)$ and $\mu(A)$ are read off $A^\top A$.

(a) (8 points) Show that for every $1 \le k \le p$,

$$\delta_k(A) \le \delta_1(A) + (k-1)\,\mu(A).$$

(b) (8 points) Let $X \in \mathbb{R}^{n\times p}$ have i.i.d. $\mathcal{N}(0,1)$ entries and $A = \frac{1}{\sqrt{n}}X$. Show that there are absolute constants $C, C' > 0$ such that if $n \ge C' \log p$, then with probability at least $1 - 2/p$,

$$\delta_1(A) \le C\sqrt{\frac{\log p}{n}} \qquad \text{and} \qquad \mu(A) \le C\sqrt{\frac{\log p}{n}}.$$

(c) (4 points) Conclude that A is k-RIP with constant δ with probability at least $1 - 2/p$ as soon as $n \ge C''\,\delta^{-2}k^2 \log p$. Note that the net argument in the lecture needs only $n \ge C\,\delta^{-2}k \log \frac{ep}{k}$ for the same conclusion. Where is the factor k lost in part (a), and why does the net argument not lose it?

3. (Vershynin 5.19, 10 points) Let X and Y be $n \times n$ symmetric matrices.

(a) (6 points) Show that if the matrices commute, i.e., $XY = YX$, then $e^{X+Y} = e^X e^Y$.

(b) (4 points) Find an example of matrices X and Y such that $e^{X+Y} \neq e^X e^Y$.

4. (The LASSO beyond the Gaussian design, 50 points) The LASSO analysis in the lecture covers designs with i.i.d. isotropic sub-Gaussian rows. In this problem you will examine the behavior of the LASSO under various design scenarios, including sub-exponential and heavy-tailed entries, correlated rows, and designs of low rank. Generate data as in Homework 3 except for the design: β^* is supported on a uniformly random set S of size k with entries ± 1, $\epsilon \sim \mathcal{N}(0, I_n)$, $y = X\beta^* + \epsilon$, $\lambda_0 = \sqrt{2\log p/n}$ and $\lambda = \frac{1}{2}\lambda_0$. Average every reported quantity over independent trials. We consider the following cases for the design X:

(A) *Gaussian*: i.i.d. $\mathcal{N}(0,1)$ entries (the baseline, as in Homework 3).

(B) *Laplace*: i.i.d. entries with density $\frac{1}{\sqrt{2}}e^{-\sqrt{2}|x|}$; sub-exponential but not sub-Gaussian.

(C) *AR(1)*: i.i.d. rows $x \sim \mathcal{N}(0,\Sigma)$ with $\Sigma_{jl} = \rho^{|j-l|}$, $\rho = 0.5$. (Generate $x_1 = z_1$ and $x_j = \rho x_{j-1} + \sqrt{1-\rho^2}\, z_j$ for $j \geq 2$, with $z_1, \ldots, z_p$ i.i.d. $\mathcal{N}(0,1)$.)

(D) *Equicorrelated*: i.i.d. rows $x = \sqrt{1-\rho}\, z + \sqrt{\rho}\, g\mathbf{1}$ with $z \sim \mathcal{N}(0, I_p)$ and $g \sim \mathcal{N}(0,1)$ independent, so $\Sigma = (1-\rho)I_p + \rho\mathbf{1}\mathbf{1}^\top$; $\rho = 0.5$ except in part (c).

(E) *Heavy-tailed*: i.i.d. entries $t/\sqrt{11}$ with $t \sim t_{2.2}$ (Student's t with 2.2 degrees of freedom). The variance is 1 but no moment of order 2.2 or higher exists.

(F) *Duplicated column* (part (b) only): design (A) with column 1 replaced by a copy of column 2; the support S always contains 2 and never 1.

(G) *Factor model* (part (b) only): rows $x = Vg$ with $g \sim \mathcal{N}(0, I_r)$, $r = 5$, where $V \in \mathbb{R}^{p\times r}$ is a fixed matrix with i.i.d. $\mathcal{N}(0,1)$ entries whose rows are rescaled to unit length (draw a new V in each trial). Then $\Sigma = VV^\top$ has rank $r < k$.

(a) (Restricted isometry constants, 20 points) Let $p = 64$, $k \in \{1,2,3,4\}$ and $n \in \{20, 40, 80, 160, 320, 640\}$, for the designs (A)–(E). Compute $\delta_k(\frac{1}{\sqrt{n}}X) = \max_{|T|=k} \|\frac{1}{n}X_T^\top X_T - I_k\|_{\text{op}}$ *exactly*, by enumerating all $\binom{64}{k} \leq 635{,}376$ supports. Average over 10 trials. Plot δ_k against n on log-log axes, one panel per k, one curve per design, with a reference line of slope $-\frac{1}{2}$ (format: Figure 1; 8 points). Then discuss:

i. (3 points) For which designs does δ_k decay like $n^{-1/2}$, as is predicted for sub-Gaussian designs? Does the prediction still hold for the non-sub-Gaussian design (B)?

ii. (5 points) Describe how δ_k behaves for (C) and (D) as n grows, and explain it through the limit of $\frac{1}{n}X_T^\top X_T$ as $n \to \infty$; compute $\lim_{n\to\infty}\delta_k$ in closed form for (D) and compare with your plot. Can a correlated design satisfy k-RIP with a small constant for large n?

iii. (4 points) Describe how δ_k behaves for (E), including $k = 1$ (the column norms), and explain it. Which assumption in the lecture's proof of restricted isometry for random designs fails?

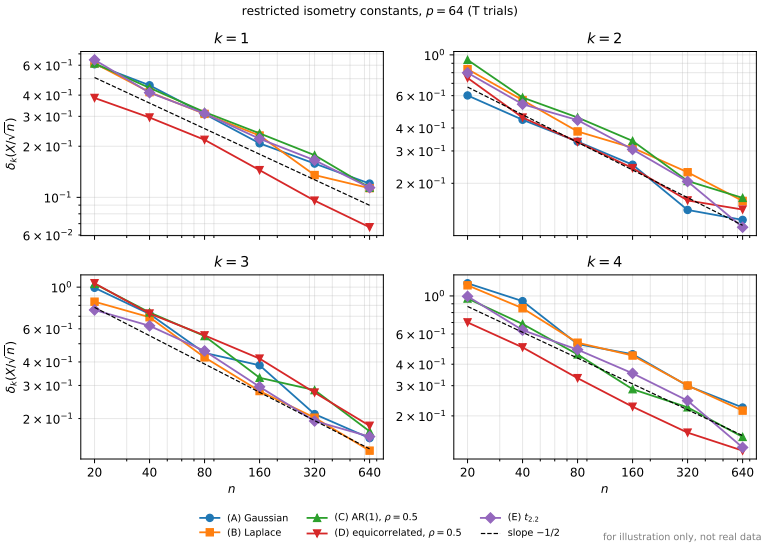


Figure 1: Expected format of the figure for part (a), for illustration only (not real data).

(b) (LASSO recovery, 20 points) Let $p = 512$, $k = 10$ and $n \in \{200, 400, 800, 1600, 3200\}$, for the designs (A), (D), (E), (F) and (G), with at least 20 trials. Record both the estimation error $\|\hat{\beta} - \beta^*\|_2^2$ and the prediction error $\frac{1}{n}\|X(\hat{\beta} - \beta^*)\|_2^2$, and plot their averages against n in two panels, on log-log axes, one curve per design, with a reference line of slope -1 (format: Figure 2; for (A) you may reuse your Homework 3 results, which used the same p, k, n and λ). For (A) and (E), also report $\mathbb{E}\|\frac{1}{n}X^\top \epsilon\|_\infty / \lambda_0$ for each n, as in Homework 3. (Figure and table: 6 points.) Then discuss:

i. (5 points) Compare what part (a) says about k-RIP for (D) with the rate of the LASSO error for (D). Explain through the restricted eigenvalue condition: show that $u^\top \Sigma u \geq \lambda_{\min}(\Sigma)\|u\|_2^2$ for every $u \in \mathbb{R}^p$, compute $\lambda_{\min}(\Sigma)$ for (D), and argue that the population version of (RE) holds with $\kappa = \lambda_{\min}(\Sigma)$. Which side of the restricted isometry inequality does the LASSO analysis use?

ii. (4 points) For the heavy-tailed design (E), compare the estimation error and the prediction error with those of (A). The bounds of Problem 1 rest on two conditions, the noise condition (N) and the design condition (RE), and $\lambda = \frac{1}{2}\lambda_0$ is the same for every design. Which of the two conditions do the heavy tails affect? Support your answer with the noise levels $\mathbb{E}\|\frac{1}{n}X^\top \epsilon\|_\infty / \lambda_0$ you reported for (A) and (E), and with what the two panels show.

iii. (5 points) For (F), exhibit a vector $\Delta \neq 0$ in the cone $\|\Delta_{S^c}\|_1 \leq 3\|\Delta_S\|_1$ with $X\Delta = 0$, so that (RE) fails. For (G), show that such a Δ exists as well, supported on S: what is the rank of X, and what does it say about the columns X_S? What does this imply for the estimation error as $n \to \infty$, and what about the prediction error? For (F), what does your solver return for the coordinates 1 and 2 of $\hat{\beta}$, and why? (The minimizer is not unique; say which solver you used.)

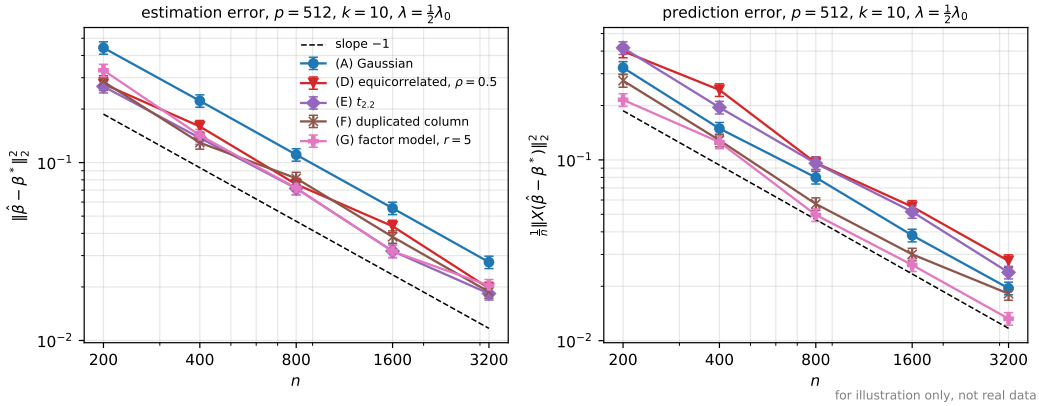


Figure 2: Expected format of the figure for part (b), for illustration only (not real data). Left: estimation error. Right: prediction error. Error bars are one standard error of the mean.

(c) (Slow versus fast rate, 10 points) Take the equicorrelated design (D) with $p = 512$, $k = 10$, $n = 800$, at least 20 trials, and $\rho \in \{0, 0.5, 0.8, 0.9, 0.95, 0.98, 0.99\}$. Plot against ρ, on a logarithmic vertical axis, the estimation error $\mathbb{E}\|\hat{\beta} - \beta^*\|_2^2$, the prediction error $\mathbb{E}\frac{1}{n}\|X(\hat{\beta} - \beta^*)\|_2^2$, and the two prediction bounds of Problem 1: $9k\lambda^2/\kappa$ with $\kappa = 1 - \rho$ (Problem 1(c)) and $3\lambda\|\beta^*\|_1$ (Problem 1(d)); format: Figure 3; 4 points. Then (2 points each): how do the two errors behave as $\rho \to 1$? Is each of the two bounds respected for

every ρ, and which one is informative for ρ close to 1? Explain the difference between the two errors for ρ close to 1.

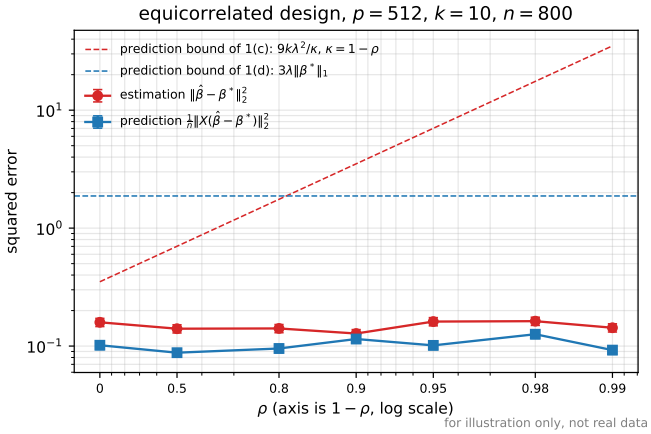


Figure 3: Expected format of the figure for part (c), for illustration only (not real data).